

School of Computing Science and Engineering

Bachelor of Technology in Computer Science and Engineering

Semester End Examination - Feb 2024

Duration : 180 Minutes

Max Marks : 100

Sem III - E2UC303B - Data StructuresGeneral Instructions

Answer to the specific question asked

Draw neat, labelled diagrams wherever necessary

Approved data hand books are allowed subject to verification by the Invigilator

- 1) Which sorting algorithm will you use to sort an almost sorted array and why? K1 (2)
- 2) The height of a tree is the length of the longest "root-to-leaf" path in it. Calculate the maximum and the minimum number of nodes in a binary tree of height 4. K2 (4)
- 3) Write a recursive and non-recursive methods in Java for binary search. K2 (6)
- 4) Write a program to implement the improved version of Bubble Sort in such a way that if numbers are already sorted then skip the further steps. K3 (9)
- 5) What is Stack? Write an algorithm or program for the linked list implementation of stack. K3 (9)
- 6) The keys 112, 117, 313, 2, 55, 43, 5 and 25 are inserted into an initially empty hash table of length 15 using open addressing with hash function $h(k) = k \bmod 10$ and linear probing. What is the resultant hash table? K5 (10)
- 7) Write an algorithm/pseudocode for Heap Sort. Use Heap sort algorithm, sort the following sequence: 18, 25, 45, 34, 36, 51, 43, and 24. K4 (12)
- 8) Write a program to find saddle point of a matrix. A saddle point is an element of the matrix such that it is the minimum element in its row and the maximum in its column. K5 (15)
- 9) Write an algorithm for converting an infix expression into postfix expression. Convert following infix expression into its equivalent postfix expression:
 $A + (B * C - (D / E \wedge F) * G) * H$ K5 (15)
- 10) Write a Java program to print a square matrix in spiral form. K6 (18)

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Sem III - E2UC303B - Data StructuresGeneral Instructions

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- 1) The minimum number of comparisons required to determine if an integer appears more than $(n/2)$ times in a sorted array of 'n' integers is _____. K1 (1)
- 2) Suppose an array, $A[-10 \dots +2]$ having Base address (BA) = 999 and size of an element = 2 bytes, find the location of $A[-1]$. K2 (2)
- 3) Write a Java program to calculate the average value of array elements. K3 (3)
- 4) Write a pseudocode or Java program to find the maximum and minimum value of an array. K3 (6)
- 5) Suppose a 3-D array A is declared using $A[1:10, -5:5, -10:5]$. (i) Find the length of each dimension and the number of elements in A. (ii) Explain Row-Major order and Column-Major order in detail with explanation formula expression. K3 (9)
- 6) Define the following terms with suitable examples: (i) Time complexity, (ii) Space Complexity, (iii) Asymptotic Notations, (iv) Big O Notations. K4 (8)
- 7) Solve a Tower of Hanoi problem for n disks and 5 towers. K5 (15)
- 8) Write algorithms of Insertion sort. Implement the same on the given numbers: 13, 16, 10, 11, 4, 12, 6, 7. K6 (6)

School of Computing Science and Engineering
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 Semester End Examination - Jan 2024

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Sem III - E2UC304C - Java and JavaScript

General Instructions

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- 1) How can you create a simple AWT application with a button. Write code for it? K1 (2)
- 2) Write a java program to use "static" keyword in mentioning variable and method. Also write static block. K2 (4)
- 3) How does String class differ from the StringBuffer class? Explain with the help of suitable example K2 (6)
- 4) What is a stream in Java? Write a Java IO program to read & write in a File. K3 (9)
- 5) Create a form for student information. Write a Java script code to find total, average, result and grade. K3 (9)
- 6) Explain the role of constructor in Java. How is parametrized constructor different from default constructor? Is it necessary to define default constructor explicitly, if there is parametrized constructor present while creating object using default constructor? Explain with a suitable example. K5 (10)
- 7) Create three threads and perform following tasks: First thread generates random integer at every 1 second and if the value is even, second thread computes the square of the number and prints. If the value is odd, the third thread will print the value of cube of the number. K4 (12)
- 8) Write a Java program to read numbers from two files "file1.txt" and "file2.txt" and stores the sorted data in file3.txt. K5 (15)
- 9) Create an user interface using Swing and apply insert, delete, update operations. Also show the students details K5 (15)
- 10) Consider a banking system implemented in Java using inner classes. The Bank class has an inner class Account, and the Account class has an inner class Transaction. Implement the Bank, Account, and Transaction classes with appropriate methods and demonstrate how to create a Bank class, add accounts, and perform transactions using inner classes. K6 (18)

School of Computing Science and EngineeringBachelor of Technology in Computer Science and Engineering
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Max Marks : 50**Sem III - E2UC304C - Java and JavaScript**General Instructions

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- 1) What is the benefit of "Unicode" in Java? K1 (1)
- 2) How does Java Garbage Collection work? K2 (2)
- 3) Write down the result of the following code K3 (3)

```
class j_operator
{
    public static void main(String args[])
    {
        byte x = 64;
        int i;
        byte y;
        i = x << 2;
        y = (byte) (x << 2);
        System.out.print(i + " " + y);
    }
}
```

- 4) What is the use of finally block in exception handling. Can we write Java statements after the finally block? Explain K3 (6)
- 5) What is the purpose of inner class in Java? Create inner classes using all possible ways. K3 (9)
- 6) Write a java program to read a string through keyboard and rewrite it in the alphabetical order. For example, the word STRING should be written as GINRST. K4 (8)

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School of Computing Science and Engineering
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Duration : 180 Minutes
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Sem III - E2UC302B - Data Base Management System

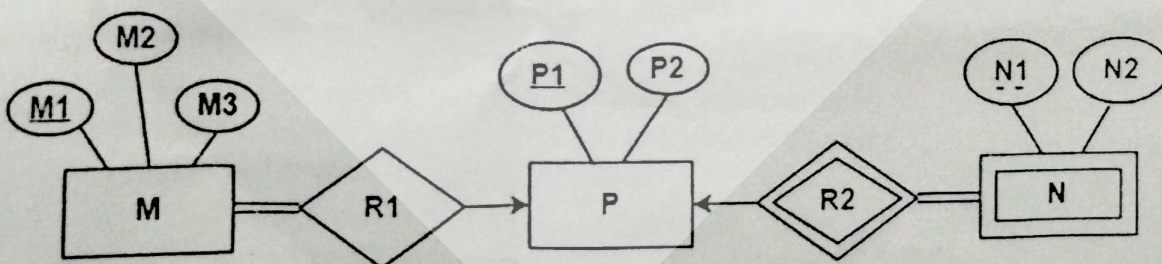
General Instructions

Answer to the specific question asked

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- 1) Why we call BCNF as the stronger version of 3NF. Explain using suitable examples. K1 (2)
- 2) Differentiate between Primary key and Foreign key with suitable examples. K2 (4)
- 3) Consider the relation Student having attributes (SID, Sname, Sdepartment, Scity) . Write SQL query for following: (a) Show the ID and name of students whoes city name starts with "P". (b) Show the ID and name of students whoes city name ends with "nagar" (c) Show the ID and name and department of students belongs to dehradun. K2 (6)
- 4) How do views contribute to data security and access control in a database system? Discuss a case. K3 (9)
- 5) Let R (E, F, G, H, I, J, K, L, M, N) a relation having a set of (FD)^{*} functional dependencies as - { E, H } → { K, L }, { E, F } → { G }, { F } → { I, J }, { L } → { N }, { K } → { M }. Calculate all the possible number of candidates keys for the given set. K3 (9)
- 6) Consider the following schema: Flights(fino, from, to, distance, departs, arrive Aircraft(aid, aname, cruisingrange) Certified (eid, aid) Employee(eid, ename, salary) Answer the following SQL queries: a) Find flights details that operate from 'Mumbai'. b) Find name of the employee who is having highest salary. c) Find employee id who are certified to fly Boeing aircraft. K5 (10)
- 7) Find the minimum number of tables required for the following ER diagram in the relational tables. Write SQL syntax to create tables, attribute types & and relationships. K4 (12)



- 8) How does functional dependencies affect the Normal form of Database design. Explain different types of functional dependencies with examples. K5 (15)

K5 (15)

K6 (18)

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School of Computing Science and Engineering
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Sem III - E2UC302B - Data Base Management System

General Instructions

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- 1) Elaborate the term RDBMS. K1 (1)
- 2) List any four significant differences between a File Management System & DBMS. K2 (2)
- 3) Consider the relation Employees having attributes (EID, Ename, Esal, Ecity) . Write SQL query to list out the ID and names of employees group by city and order by salary. K3 (3)
- 4) Consider a table Employee having columns (eID, Ename, eCity, eSal). Write SQL Syntax for the following:- (a) Add a new column named eProjects. (b) Drop the column eCity. (c) Rename the table as Employee_Info. K3 (6)
- 5) Consider the relational database of employee (person name, street, city) works (person name, company name, salary) company (company name, city). Give an expression in the relational algebra to express each of the following queries: (a) Find the names of all employees who work for "First Bank Corporation". (b) Find the names and cities of residence of all employees who work for "First Bank Corporation". (c) Find the names, street address, and cities of residence of all employees who work for "First Bank Corporation" and earn more than \$10,000. K3 (9)
- 6) Provide a brief description of normalization in relational database architecture, explaining the different levels of normalization using illustrative examples. K4 (8)
- 7) Write SQL syntax to create tables, attribute types and relationships. Find the minimum number of tables required for the ER diagram in the above relational tables. K5 (15)
- 8) Consider the relations Orders (OrderID, Orderdate, ordertype, orderitemsID) and Items (ItemID, Quantity). Write SQL statements for performing theta Join and outer join operations. K6 (6)

School of Computing Science and Engineering

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Sem III - E2UC301T - Computer Organisation and Architecture
General Instructions

Answer to the specific question asked

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- 1) Determine the number of clock cycles that it takes to process 100 tasks in a four-segment pipeline. K1 (2)
- 2) Show the hardware that implements the following RTL statement $xyTo+T1+y'T2: AR \leftarrow AR+BR$ where AR and BR is a n-bit registers and x, y, T0, T1 and T2 are control variables. K2 (4)
- 3) What is the value of output of a 4 bit combination circuit shifter, if input A is 1010, S=1. In this particular circuit, S=0 is for right shift and S=1 for left shift, Input right bit(IR)= 0, and Input Left bit(IL) = 1. How the logical shift is different from circular shift and arithmetic shift micro-operation? K2 (6)
- 4) Draw a flowchart to depict arithmetic pipeline for floating point addition/subtraction . Also, use a numerical example to explain it. K3 (9)
- 5) Apply the concept of pipelining to the instruction pipeline. Sketch space time diagram for it. K3 (9)
- 6) Represent the following signed numbers in binary using eight bits. +83, -83, +68, -68. (i) Perform the addition $(-83) + (+68)$ in binary and interpret the result obtained. (ii) Perform the subtraction $(-68) - (+83)$ in binary and interpret the result obtained. K5 (10)
- 7) Demonstrate the all the phases of instruction and show the contents in hexadecimal of registers PC, AR, DR, IR and SC of the basic computer when an ISZ indirect instruction is fetched from memory and executed. The initial content of PC is 7FF. The content of memory at address 7FF is EA 9F. The content of memory at address A9F is 0C35. The content of memory at address C35 is FFFF. K4 (12)
- 8) Defend different types of mode of transfer in I/O organization. K5 (15)
- 9) Design a flowchart for CPU program to input data device using Programmed I/O mode of transfer and explain this data transfer mode. K5 (15)
- 10) Draw the flowchart for instruction cycle and depict all the micro-operations of all phases of the instruction cycle. K6 (18)

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- 1) Determine the address bus size of a digital system having memory of specification 1024 kilobytes . K1 (1)
- 2) Explain what is wrong with the following register transfer statements:
(i) $xT:AR \leftarrow (AR), AR \leftarrow 0$ (ii) $yT:R1 \leftarrow R2, R1 \leftarrow R3$ K2 (2)
- 3) Convert the following arithmetic expressions from infix to postfix: $A + (B * C - (D / E ^ F) * G) * H$ K3 (3)
- 4) Convert the following numerical arithmetic expressions into reverse polish notation: $(3+4)[10(2+6)+8]$ and show the stack operations for evaluating the numerical result. K3 (6)
- 5) Design a 4- bit full adder using two half adders. K3 (9)
- 6) A computer uses a memory unit with 1024 K words of 64 bits each. A binary instruction code is stored in one word of memory. The instruction has four parts: an indirect bit, an operation code, A register code part to specify one of 128 registers and an address part. Draw the instruction word format and indicate the number of bits in each part. K4 (8)
- 7) Show the contents of registers E, A, Q and SC during the process of multiplication of two binary numbers 11111 (multiplicand) and 10101 (multiplier). The signs are not included. K5 (15)
- 8) What is a bus? Design a bus system capable of transmitting data from any register of a group of 8 registers (32-bits each) to any other register in a group of 8 registers (32-bits each). Illustrate the logic through its block diagram? K6 (6)

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Sem III - C1UC322T - Probability and Statistics
General Instructions

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- 1) Define Type one Error and Type Two Error. K1 (3)
- 2) In a certain industrial facility, accidents occur infrequently. It is known that the probability of an accident on any given day is 0.005 and accidents are independent of each other (a) What is the probability that in any given period of 400 days there will be an accident on one day? (b) What is the probability that there are at most three days with an accident? K2 (4)
- 3) Find rank correlation coefficient of following data K2 (6)

x	60	34	40	50	45	41	22	43	42	66	64	46
y	75	32	34	40	45	33	12	30	36	72	41	57
- 4) An urn contains four balls, each ball having equal probability of being white or black. Three black balls are added to the urn. What is the probability that five balls in the urn are black? K3 (6)
- 5) A random sample of 100 recorded deaths in the United States during the past year showed an average life span of 71.8 years. Assuming a population standard deviation of 8.9 years, does this seem to indicate that the mean life span today is greater than 70 years? Use a 0.05 level of significance. K3 (6)
- 6) Let the random variable X represent the number of automobiles that are used for official business purposes on any work day. The probability distribution of company A is K3 (9)

x	1	2	3
f(x)	0.3	0.4	0.3

And that for company B is

X	0	1	2	3	4
F(x)	0.2	0.1	0.3	0.3	0.1

Show that the variance of the probability distribution for company B is greater than that for company A. K3 (9)
- 7) The distribution function for a random variable X is K3 (9)

$$F(x) = \begin{cases} 1 - e^{-2x}, & x \geq 0 \\ 0, & x < 0 \end{cases}$$

(a) Find density function (b) Probability that $X > 2$ and (c) The probability that $-3 < X \leq 4$.

- 8) Compute Mean, Mode and standard deviation from following data

K4 (8)

x	6	7	8	9	10	11	12
y	3	6	9	13	8	5	4

- 9) Find the value of c when probability density function of a random variable X is

K4 (12)

$$f(x) = \begin{cases} x, & 0 \leq x < \frac{1}{2} \\ c(2x-1)^2, & \frac{1}{2} < x \leq 1 \\ 0, & \text{otherwise} \end{cases}$$

- 10) Fit a straight line to the following data

K5 (10)

x	1	2	3	4	5
y	14	27	40	55	68

- 11) The joint probability function of two discrete random variables X and Y is given by $f(x,y) = c(2x+y)$, where x and y can assume all integers such that

K5 (15)

$0 \leq x \leq 2, 0 \leq y \leq 3$ and $f(x,y)=0$ otherwise.

1. Find the value of the constant c,

2. Find $P(X \geq 1, Y \leq 2)$.

OR

If the density function of a continuous random variable X is given by

K5 (15)

$$f(x) = \begin{cases} 0, & x < 0 \\ ax, & 0 \leq x \leq 2 \\ (4-x)a, & 2 \leq x \leq 4 \\ 0, & x > 4 \end{cases}$$

(i) Find the value of a. (ii) Find cdf of X.

- 12) Suppose that the error in the reaction temperature in degree centigrade for a Laboratory experiment is a continuous random variable X having the probability density function

K6 (12)

$$f(x) = \begin{cases} x^2/3, & -1 < x < 2 \\ 0, & \text{elsewhere} \end{cases}$$

1. Verify that $f(x)$ is a density function.

2. Find $P(0 < X < 1)$.

OR

The television picture tubes of manufacturer A have a mean lifetime of 6.5 years and a standard deviation of 0.9 year, while those of manufacturer B have a mean lifetime of 6.0 years and a standard deviation of 0.8 year. What is the probability that a random sample of 36 tubes from manufacturer A will have a mean lifetime that is at least 1 year more than the mean lifetime of a sample of 49 tubes from manufacturer B?

K6 (12)

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Sem III - C1UC322T - Probability and Statistics

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- 1) State Baye's theorem of probability. K2 (2)
- 2) If A,B,C are mutually exclusive events associated with a random experiment. $P(B)=0.6P(A)$, $P(C)=0.6P(A)$, then find $P(A)$. K1 (3)
- 3) From the following data compute standard deviation K2 (4)

Mass in kg	60-62	63-65	66-68	69-71	72-74
No of students	5	18	42	27	8
- 4) Compute Spearman's rank correlation coefficient from following data K2 (6)

Person	A	B	C	D	E	F	G	H	I	J
Rank in Stat	9	10	6	5	7	2	4	8	1	3
Rank in Economics	1	2	3	4	5	6	7	8	9	10
- 5) Find the mode from the following data. K3 (6)

Age	0-6	6-12	12-18	18-24	24-30	30-36	36-42
Frequency	6	11	25	35	18	12	6
- 6) From the following data calculate Karl-Pearson's coefficient of correlation. K3 (9)

x	1	3	5	7	8	10
y	8	12	15	17	18	20
- 7) Find mean and standard deviation for following data K4 (8)

Age in yrs	20-30	30-40	40-50	50-60	60-70	70-80	80-90
No of members	3	61	132	153	140	51	2
- 8) A dice is tossed twice and sum of numbers appearing is noted to be 8. What is the conditional probability that the number 5 has appeared at least once. K4 (12)

OR

Find coefficient of correlation when lines of regression are $2x-9y+6=0$ and $x-2y+1=0$ K4 (12)

$$2y = x + 1$$

$$y = \frac{x}{2} + \frac{1}{2}$$

$$\frac{1}{2} \times \frac{2}{9} = \frac{1}{9}$$

$$9y = 2x + 6$$

$$y = \frac{2}{9}x + \frac{6}{9}$$